

POS	PLAYER	MAT	INNS	NO	RUNS	HS	AVG	BF	SR	100	50	4S	6S
1	 Virat Kohli RCB	15	15	3	741	113*	61.75	479	154.69	1	5	62	38
2	 Ruturaj Gaikwad CSK	14	14	3	583	108*	53.00	413	141.16	1	4	58	18
3	 Riyan Parag RR	16	14	3	573	84*	52.09	384	149.21	0	4	40	33
5	 Sanju Samson RR	16	15	4	531	86	48.27	346	153.46	0	5	48	24
6	 Sai Sudharsan GT	12	12	1	527	103	47.91	373	141.28	1	2	48	16
7	 K L Rahul LSG	14	14	0	520	82	37.14	382	136.12	0	4	45	19
10	 Abhishek Sharma SRH	16	16	1	484	75*	32.27	237	204.21	0	3	36	42
12	 Rishabh Pant DC	13	13	2	446	88*	40.55	287	155.40	0	3	36	25
15	 Yashasvi Jaiswal RR	16	15	1	435	104*	31.07	279	155.91	1	1	54	16
16	 Shubman Gill GT	12	12	1	426	104	38.73	289	147.40	1	2	37	15

POS	PLAYER	MAT	INNS	NO	RUNS	HS	AVG	BF	SR	100	50	4S	6S
4	 Travis Head SRH	15	15	1	567	102	40.50	296	191.55	1	4	64	32
8	 Nicholas Pooran LSG	14	14	6	499	75	62.38	280	178.21	0	3	35	36
9	 Sunil Narine KKR	15	14	0	488	109	34.86	270	180.74	1	3	50	33
11	 Heinrich Klaasen SRH	16	15	3	479	80*	39.92	280	171.07	0	4	19	38
13	 Faf Du Plessis RCB	15	15	0	438	64	29.20	271	161.62	0	4	47	21
14	 Phil Salt KKR	12	12	1	435	89*	39.55	239	182.00	0	4	50	24
21	 Marcus Stoinis LSG	14	14	2	388	124*	32.33	263	147.52	1	2	39	16
22	 Tristan Stubbs DC	14	13	6	378	71*	54.00	198	190.90	0	3	24	26
24	 Jos Buttler RR	11	11	2	359	107*	39.89	255	140.78	2	0	36	12
29	 Jake Fraser - McGurk DC	9	9	0	330	84	36.67	141	234.04	0	4	32	28

MEAN OF LOCAL PLAYERS:

For getting the mean we have take the data of highest run from each local and foreign player:

We can get the mean (average) of the highest runs scored by each local player in the IPL match, we need to sum all the scores and then divide by the number of players.

1. Sum of the highest runs: $113+108+84+86+103+82+75+88+104+104=947$
2. Number of players: 10

mean = Sum of the highest run / number of player

$$= 947/10$$

$$= 94.7$$

So, the mean of the highest runs scored by each local player is 94.7.

MEAN OF FOREIGN PLAYER

1. Sum the highest runs: $102+75+109+80+64+89+124+71+107+84= 905$
2. the number of players: 10

Mean :

=Sum of the highest number / number of player

$$= 905/10$$

$$= 90.5$$

So, the mean of the highest runs scored by each foreign local player is 90.5.

Analysis According to MEAN :

1. **Comparison of Means:**
 - The mean highest runs for local players is 94.7.
 - The mean highest runs for foreign players is 90.5.
2. **Performance Insights:**
 - The local players have a higher mean score (94.7) compared to the foreign players (90.5), indicating that on average, local players have performed slightly better in terms of their highest individual scores.

- The difference between the means is $94.7 - 90.5 = 4.2$ which means that the local player's performance is slightly better than the foreign player.

MODE of Local Players by using strike rate :

Frequency Table with Class Boundaries:

We use the strike rate value of each local player in IPL 2024 as class interval and frequency :

Class interval	frequency	Class boundary
136.12 - 149.74	5	136.115 - 149.745
149.74 - 163.36	4	149.735 - 163.365
163.36 - 176.98	0	163.355 - 176.985
176.98 - 190.60	0	176.975 - 190.605
190.60 - 204.21	1	190.595 - 204.215

The data in the table is a grouped data. To get the mode for the grouped data, we need to use the formula for the mode in a frequency distribution. The mode for grouped data is calculated using the following formula:

$$\text{Mode} : L + \left(\frac{fm - fm-1}{(fm - fm-1) + (fm - fm+1)} \right) * h$$

L is the lower boundary of the modal class.

fm is the frequency of the modal class.

fm-1 is the frequency of the class preceding the modal class.

fm+1 is the frequency of the class succeeding the modal class.

h is the class width.

Given Data:

Modal class: 136.12 - 149.74

L=136.115

fm=5 (frequency of the modal class)

fm-1=0 (frequency of the class before the modal class)

fm+1=0 (frequency of the class after the modal class)

h=13.62 (class width)

h=17.30 (class width)

$$\begin{aligned}\text{Mode} &= 136.115 + ((5-0)/(5-4) + (5-0) \times 13.62) \\ &= 36.115 + 0.8333 \times 13.62 \\ &\approx 147.4643\end{aligned}$$

The mode for the local player is 147.4643

MODE of Foreign Player by using strike rate :

Class interval	frequency	Class boundary
147.52 - 164.82	147.515 - 164.825	1
164.82 - 182.12	164.815 - 182.125	2
182.12 - 199.42	182.115 - 199.425	3
199.42 - 216.72	199.415 - 216.725	2
216.72 - 234.04	216.715 - 234.045	1

Here

L=182.115 (lower boundary of the modal class)

fm=3 (frequency of the modal class)

fm-1=2 (frequency of the class before the modal class)

fm+1=2 (frequency of the class after the modal class)

h=17.30 (class width)

Mode :

$$\begin{aligned}&= 182.115 + ((3-2)/(3-2) + (3-2)) \times 17.30 \\ &= 182.115 + 8.65 \\ &= 190.765\end{aligned}$$

Performance Analysis Based on MODE :

The mode value for local players is 147.46, whereas for foreign players, it is significantly higher at 190.77. foreign players frequently achieve their highest strike rates around 190.77. This much higher mode value indicates that foreign players are capable of delivering exceptionally high-performance innings in IPLI 2024. Therefore, based on the mode analysis, foreign players are better performers in the context of achieving high strike rates in IPL matches.

Median for local player (using highest runs):

Class interval	Class Boundary	Midpoint	Frequency	Cumulative Frequency
75 - 82	74.5 - 82.5	78.5	2	2
83 - 90	82.5 - 90.5	86.5	3	5
91 - 98	90.5 - 98.5	94.5	0	5
99 - 106	98.5 - 106.5	102.5	3	8

107 - 113	106.5 - 113.5	110.0	2	10
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MEDIAN (LOCAL PLAYER)

To get the median value for local player is

$$\text{Median} = L + (N/2 - CF/f) \times w$$

where:

- L = lower boundary of the median class
- N = total number of observations
- CF = cumulative frequency of the class preceding the median class
- f= frequency of the median class
- c = class size

Median Calculation:

Class interval: 83–90

Lower boundary L: 82.5

Cumulative frequency of the class preceding the median class CF: 2

Frequency of the median class f: 3

Class size : 7

$$\text{Median} = 82.5 + (5 - 2/3) \times 7$$

$$= 82.5 + 1 \times 7$$

$$= 89.5$$

So the median value for the indian local player is 89.5

Class Interval	Class Boundary	Midpoint	Frequency	Cumulative Frequency
64 - 75	112 - 124	69.5	3	3
76 - 87	75.5 - 87.5	81.5	2	5
88 - 99	87.5 - 99.5	93.5	1	6
100 - 111	99.5 - 111.5	105.5	3	9
112 - 124	111.5 - 124.5	118.0	1	10

Here $N = 10$

So the position of the median class is : $N/2 = 10/2 = 5$ th class that is (76-87)

Median Class:

- Class interval: 76–87
- Lower boundary L: 75.5
- Cumulative frequency of the class preceding the median class CF: 3
- Frequency of the median class f: 2
- Class size :12

$$\text{Median} = L + (N/2 - CF/f) \times c$$

$$= 75.5 + (5 - 3/2) \times 11$$

$$= 75.5 + 1 \times 11$$

$$= 86.5$$

So the median for Foreign Player IPL is 86.5

The median values for the runs scored by the local player and the foreign player are:

- **Local Player Median:** 89.5
- **Foreign Player Median:** 86.5

Analysis:

- The median run score for the local player (89.5) is higher than that for the foreign player (86.5).
- The median represents the middle value in a data set when it is ordered from least to greatest. A higher median indicates that the central tendency of the local player's performance is better than that of the foreign player.

Chapter 6

For Indian players

First, sort the average if the each Indian players in ascending order:

31.07, 32.27, 37.14, 38.73, 40.55, 47.91, 48.27, 52.09, 53.00, 61.75

Now, let's define the class intervals:

Class 1: 30.0 - 36.6

Class 2: 36.7 - 43.3

Class 3: 43.4 - 50.0

Class 4: 50.1 - 56.7

Class 5: 56.8 - 63.0

Class Interval	Frequency	Cumulative Frequency	Lower Boundary	Higher Boundary
30.0 - 36.6	2	2	29.5	36.65
36.7 - 43.3	3	5	36.65	43.35
43.4 - 50.0	2	7	43.35	50.05
50.1 - 56.7	2	9	50.05	56.75
56.8 - 63.0	1	10	56.75	63.05

Where:

L = lower boundary of the median class

n = total number of frequencies

F = cumulative frequency of the class before the median class

f = frequency of the median class

w = class width

Values for the Median Class:

$$L = 36.65$$

$$N = 10$$

$$F = 2 \text{ (cumulative frequency of the class before the median class)}$$

$$f = 3 \text{ (frequency of the median class)}$$

$$W = 43.35 - 36.65 = 6.7 \text{ (class width)}$$

Substituting the values into the formula:

$$\text{Median} = 36.65 + \left(\frac{(10/2) - 2}{3} \right) \times 6.7$$

$$\text{Median} = 36.65 + \left(\frac{3}{3} \right) \times 6.7$$

$$\text{Median} = 36.65 + 1 \times 6.7$$

$$\text{Median} = 36.65 + 6.7 = 43.35$$

So, the median batting average is **43.35**.

For Foreign Players

First, sort the average if the each Overseas players in ascending order:

29.20, 32.33, 34.86, 36.67, 39.55, 39.89, 39.92, 40.50, 54.00, 62.38

Now, let's define the class intervals:

Class 1: 29.1 - 35.8

Class 2: 35.9 - 42.6

Class 3: 42.7 - 49.4

Class 4: 49.5 - 56.2

Class 5: 56.3 - 63.0

Class Interval	Frequency	Cumulative Frequency	Lower Boundary	Upper Boundary
29.1 - 35.8	3	3	29.05	35.85
35.9 - 42.6	5	8	35.85	42.65
42.7 - 49.4	0	8	42.65	49.45
49.5 - 56.2	1	9	49.45	56.25
56.3 - 63.0	1	10	56.25	63.05

Where:

L = lower boundary of the median class

n = total number of frequencies

F = cumulative frequency of the class before the median class

f = frequency of the median class

w = class width

Values for the Median Class:

$$L = 35.85$$

$$N = 10$$

$$F = 3 \text{ (cumulative frequency of the class before the median class)}$$

$$f = 5 \text{ (frequency of the median class)}$$

$$W = 35.85 - 29.05 = 6.8 \text{ (class width)}$$

Substituting the values into the formula:

$$\text{Median} = 35.85 + \left(\left(\frac{10}{2} \right) - \frac{2}{5} \right) \times 6.8$$

$$\text{Median} = 35.85 + \left(\frac{2}{5} \right) \times 6.8$$

$$\text{Median} = 35.85 + 0.4 \times 6.8$$

$$\text{Median} = 35.85 + 2.72 = 38.57$$

So, the median batting average is **38.57**

Chapter 01

1. Probability of a player having an average above 40

Indian Players:

- Total Indian players: 10
- Players with an average above 40: Virat Kohli (61.75), Ruturaj Gaikwad (53.00), Sanju Samson (48.27), Rishabh Pant (40.55) = 4
- Probability: $\frac{4}{10} = 0.4$ or 40%

Overseas Players:

- Total overseas players: 10

- Players with an average above 40: Travis Head (40.50), Nicholas Pooran (62.38), Heinrich Klaasen (39.92), Phil Salt (39.55), Jos Buttler (39.89) = 2
- Probability: $\frac{2}{10} = 0.2$ or 20%

2. Probability of a player hitting more than 30 sixes

Indian Players:

- Total Indian players: 10
- Players with more than 30 sixes: Virat Kohli (38), Riyan Parag (33), Sanju Samson (24), Abhishek Sharma (42), Rishabh Pant (25) = 3
- Probability: $\frac{3}{10} = 0.3$ or 30%

Overseas Players:

- Total overseas players: 10
- Players with more than 30 sixes: Travis Head (32), Nicholas Pooran (36), Sunil Narine (33), Heinrich Klaasen (38), Jake Fraser-McGurk (28) = 4
- Probability: $\frac{4}{10} = 0.4$ or 40%

3. Probability of a player having a strike rate above 150

Indian Players:

- Total Indian players: 10
- Players with SR > 150: Virat Kohli (154.69), Sanju Samson (153.46), Abhishek Sharma (204.21), Rishabh Pant (155.40), Yashasvi Jaiswal (155.91), Shubman Gill (147.40) = 5
- Probability: $\frac{5}{10} = 0.5$ or 50%

Overseas Players:

- Total overseas players: 10
- Players with SR > 150: Travis Head (191.55), Nicholas Pooran (178.21), Sunil Narine (180.74), Heinrich Klaasen (171.07), Phil Salt (182.00), Tristan Stubbs (190.90), Jake Fraser-McGurk (234.04) = 7
- Probability: $\frac{7}{10} = 0.7$ or 70%

4. Probability of a player scoring at least 500 runs

Indian Players:

- Total Indian players: 10
- Players with at least 500 runs: Virat Kohli (741), Ruturaj Gaikwad (583), Riyan Parag (573), Sanju Samson (531), Sai Sudharsan (527), K L Rahul (520) = 6
- Probability: $\frac{6}{10} = 0.6$ or 60%

Overseas Players:

- Total overseas players: 10
- Players with at least 500 runs: Travis Head (567) = 1
- Probability: $1 / 10 = 0.1$ or 10%